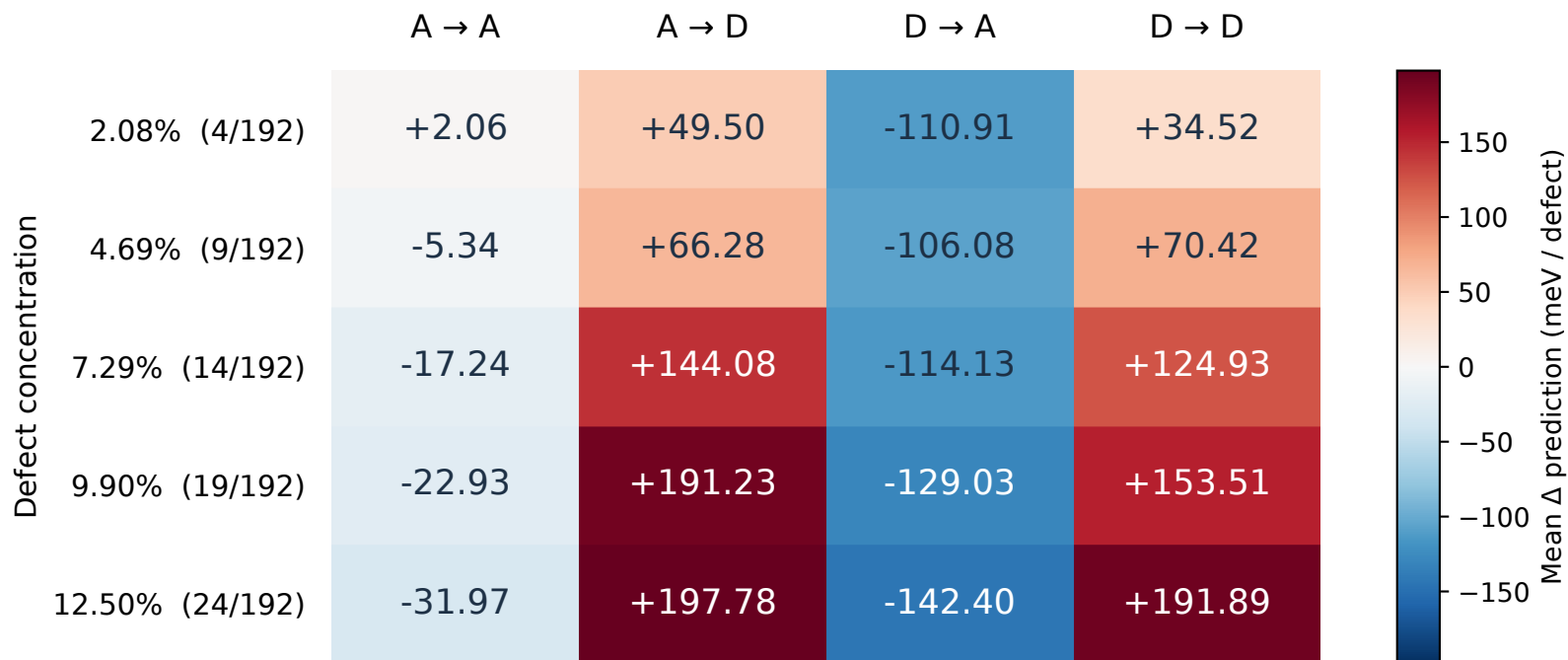


Mean relation-message intervention by concentration

Shared hetero · mixed-low checkpoint · 100 structures per concentration / material

MoS₂



A = ordinary atom; D = actual defect. Δ = prediction after intervention – original prediction.

Zero one relation's node-message numerator at every layer; gates, edge updates and geometry remain.